

Table 1: DATE ADD

SELECT: emp-id,
hire-date

DATE ADD (MONTH, 6, hire-date) AS hire-Plus-6-Months.
FROM Employees

emp-id	Name	hire-date	hire-Plus-6-Months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

SELECT: Students,
Student-id,
Name

DATE IFF (DAY, Current-Date, dob) AS age-in-days
FROM Students;

Student-id	Name	dob	age-in-days
101	Maya	2005-08-10	7557
102	Ethan	2004-12-01	7809
103	Sienna	2006-03-15	7340

Table 3: Events

SELECT event-id,
event-name,
Current-Date ()

DATE IFF (DAY, event-date, Current-Date) AS days-remaining
FROM Events;

event-id	Event-name	days-remaining
1	Seminar	-673
2	Workshop	-230
3	Hackathon	-454

TABLE 4: EMPLOYEES INVOICES

SELECT invoice-id,
issue-id,
due-date,

DATEIFF (Day, due-date, issue-date) AS days-between
FROM Invoices;

Invoice-id	Issue-date	Due-date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

SELECT Course-id,
name,

TO-CHAR (start-date, 'yyyy-mm') AS formatted-date
FROM Courses;

Course-id	Name	Formatted date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

SELECT member-id,

DATE-FROM-PARTS (start-year, start-month, start-day) AS full-start-date
FROM Memberships;

Member-id	Full-start-date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

SELECT Sub-id,
Plan,

DATEADD (YEAR, 1, renewal-date) AS extended-renewal-date
FROM Subscriptions;

Sub-id	Plan	extended-renewal-date
11	Basic	2026-01-01
12	Premium	2026-03-15

NULL FUNCTIONS

TABLE 1: Employees

SELECT employee-id,
Name,
IFNULL (Salary, 0) AS Salary-with-default
FROM Employees;

Employee-id	Name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

QUESTION 2

SELECT employee-id,
Name,
IFNULL (department, Not Assigned) AS department-name
FROM Employees;

Employee-id	Name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2: Orders

SELECT order-id,
Customer-id
FROM Orders
WHERE Customer-id IS NULL;

Order-id	Customer-id
103	NULL

QUESTION 4:

SELECT order-id,
Customer-id,
IF NULL (delivery-date, Pending) AS delivery-status.
FROM Orders;

order-id	Customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: StudentsQUESTION 5

SELECT Student-id,
name,
IF NULL (grade, 0) AS final-grade
FROM Students;

Student-id	Name	Grade
1	Ethan	85
2	Maya	0
3	Olivia	90

QUESTION 6

SELECT COUNT(*) AS not-graded-count
FROM Student
WHERE grade is NULL;
Outcome = 1

Not-graded-count
1

Table 4: ProductsQUESTION 7

SELECT Product-id,
Name,
Price,
IF NULL (discount, 0) AS final-price
FROM Products;

Product-id	Name	Price	final-price
501	keyboard	25	0
502	Mouse	15	5
503	Monitor	100	0

Table 5: CustomersQUESTION 8

SELECT COUNT(*) AS missing-email-count
FROM Customers
WHERE email is NULL;

Outcome = 2

QUESTION 9

SELECT Customer-id
name,
IF NULL (email, 'No Email') AS email-display
FROM Customers;

Customer-id	Name	email-display
1	LINDA	No display
2	JOSEPH	Joseph@gmail.com
3	NIR	No Display

Table 6: Payments

Payment-id	Method display	Status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

Table 7: Inventory

QUESTION 11

SELECT item-id,
item-name,
IF NULL (Quantity, 0) AS Quantity-Checked
FROM Inventory;

Item-id	Item-name	Quantity-Checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees Extra

QUESTION 12

SELECT emp-id,
COALESCE (bonus, commission) AS first-available-reward
FROM Employees_Extra;

emp-id	first-available-reward
1	300
2	100
3	NULL

Table 9: Classes

QUESTION 13

SELECT COUNT(*) AS no-room-count
FROM Classes
WHERE room IS NULL;

no-room-count
2

Table 10: Attendance

Outcome = 2

QUESTION 14

SELECT Student-id,
date,
IF NULL (status, 'Not Marked') AS attendance-status
FROM Attendance;

Student-id	date	attendance-status
1	2025-04-01	Not Marked
2	2024-04-01	Present
3	2025-04-01	Absent

Table 11: Bank-Accounts.Question 15

SELECT

account-id,
 IFNULL(account-type, 'Unknown') AS type-display,
 IFNULL(balance, 0) AS balance-checked
 FROM Bank-Accounts;

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Question 16Table 12: Projects

SELECT

Project-id
 title,
 IFNULL(start-date, 'TBD') AS start-display

FROM Project;

Project-id	title	Start display
1	Website Revamp	2025-01-10
2	Mobile APP	TBD
3	Data Migration	TBD

Question 17Table 13: Reviews.

SELECT

review-id,
 Product-id,
 IFNULL(comment, 'No comment') AS comment-display,
 IFNULL(rating, 0) AS rating-display

FROM Reviews;

review-id	Product-id	comment-display	rating-display
1	S01	Great-Product	4
2	S02	No comment	0
3	S03	Works fine	3

Question 18Table 14: Suppliers

SELECT

Suppliers-id,
 Name,
 COALESCE(phone-alt-phone, 'No Contact') AS Contact-number

FROM Suppliers;

Supplier-id	name	Contact number
1	Global Goods	123456789
2	Best-Supplier	987654321
3	ValueSource	No Contact